

Do the claimed waves of molecular aging survive rigorous statistical testing?

Reproducing and extending the LOESS?DE-SWAN artifact analysis

1. Executive summary

Research question. Are the widely-cited discrete transitions in human molecular aging at

Key finding. They are artifacts, and we can now say by how much. Across all ten omics layers

What is new relative to the existing critique. A June 2026 bioRxiv preprint by Carbonneau,

Two findings that go beyond the existing critique and matter most.

1. The published robustness argument has no discriminating power. Shen et al. support the ~44

Practical implication. A DE-SWAN curve computed on smoothed data is not interpretable, and no

2. Research question, hypothesis, and what would refute it

2.1 The hypothesis

> The majority of reported discrete aging transitions in omics data are statistical artifacts of

2.2 Three claims the literature conflates

The hypothesis is only testable if three claims that are routinely run together are separated,

| | Claim | Strength |

We report on each separately throughout.

2.3 Why it matters

One pipeline underwrites a fast-growing literature. Lehallier et al. 2019 (Nat Med, ~830

2.4 Pre-registered falsification conditions

Stated in planning.md before any experiment was run:

- Supports the hypothesis ? crest location tracks the age distribution and moves under

3. Data and methods

3.1 Data

| Dataset | Size | Role | Provenance |

The iPOP recovery is non-obvious and worth recording: Shen et al.'s data-availability statement

Cross-sectional construction follows the published Methods: keep healthy visits, average each

3.2 Reimplementing the pipeline

There is no R in this environment, and the pipeline needs R's loess (degree 2, tricube);

- src/fastloess.py ? R-compatible LOESS with leave-one-out span selection over

The performance work was necessary rather than cosmetic. A local polynomial fit is linear in y,

Validation (src/validate_fastloess.py, all assertions pass):

| Check | Result |

Residual disagreement with R is R's surface="interpolate" kd-tree approximation plus CV span

A second optimisation ? an argsort-based Mann-Whitney used only inside permutation loops ? is

3.3 The experiments

| | Experiment | Question |

3.4 The corrected estimator

Carbonneau et al. recommend but do not supply a replacement. src/corrected.py implements one

- Any age association? Linear F-test.

All models share one design, so every molecule is fitted in a single least-squares solve

3.5 Reproducibility

Seed 42 throughout (src/common.py). Python 3.12.11, NumPy 2.5.2, SciPy 1.18.0, pandas 3.0.5,

4. Results

4.1 E1 ? The artifact reproduces in all ten layers

![Reproduction across all ten iPOP omics layers](figures/fig1_reproduction_all_layers.png)

*Figure 1. Blue: the published LOESS + DE-SWAN pipeline. Orange: the identical Mann-Whitney test,

| Layer | Variables | Crest age | Peak % significant | Floor % significant | DE-SWAN only: significant tests / total |

Two things stand out. First, the transcriptome and proteomics crests land at 43 and 44,

The comparison arm is decisive. The same test, same multiplicity correction, same windows, applied

4.2 E2 ? Which step manufactures the signal

Two mechanisms are conflated in LOESS + DE-SWAN and had not been separated: smoothing (fitted

![Decomposition and crest instability](figures/fig2_decomposition.png)

*Figure 2. (a) Peak % of variables significant under each combination. (b) Every crest age obtained

Peak % of variables significant:

| Layer | raw (neither) | interpolate (densify only) | LOESS at observed ages (smooth only) | LOESS on grid (published) |

Smoothing is the culprit, not densification. LOESS evaluated at the observed ages ? no

Three further ablations:

- The test statistic is irrelevant. Replacing Mann-Whitney with Lehallier's linear model gives

Reported for the first time alongside a DE-SWAN curve: per-window group sizes. On the raw data

4.3 E3 ? The permutation null decides everything, and the ordering decides the null

This is the crux. Shen et al. report a permutation test under which their peaks disappear;

| | Procedure | Valid? |

We ran both, on the same data, in the same code.

![Observed curves against both permutation nulls](figures/fig3_permutation_nulls.png)

*Figure 3. Blue: observed. Green band and dashed line: the valid null (shuffle before smoothing),

| Layer | Observed peak | Valid-null mean peak | p (valid) | Invalid-null mean peak | p (invalid) | Achieved FDP |

*(300 permutations for the transcriptome, 1,000 for the rest; p-values are max-statistic

Three readings:

1. Under the valid null, nothing is significant. In four of six layers the observed peak is

![Achieved false discovery proportion](figures/fig7_achieved_fdr.png)

Where do artifactual crests land? The valid-null crest distribution has a median of 41-44 in

4.4 E4 ? Both published ages arise from data with no age relationship

![Null simulations and the power confound](figures/fig4_null_simulations.png)

(a) Pure noise reproduces the published structure. 1,000 molecules drawn i.i.d. $N(0,1)$,

Both published ages are therefore reproduced by data containing no age relationship at all ? but

(b) The curve shape is set by the age distribution alone. Identical null generator, four age

| Age distribution | Apparent crest age (null data) |

(c) DE-SWAN turns strictly linear data into waves, with no LOESS involved. Data generated with

| Scenario | Apparent crest | Peak count / 1000 | corr(counts, harmonic-mean window n) |

The counts go from 0 at the edges to a sharp peak at 54-60 on data whose truth is a straight line

(d) Resampling the real data. Subsampling iPOP subjects to 70 % while flattening the age

4.5 E8 ? The published robustness analysis cannot discriminate

Shen et al. defend the ~44 crest by showing it is stable across bucket widths {15, 20, 25, 30} and

That argument is evidence for biology **only if an artifactual crest would be unstable across the

| Layer | Real-data crest spread across the 16 cells | Median null-data spread | Fraction of null simulations at least as stable a

In proteomics the null is more robust than the real data. Stability across bucket width and

4.6 The mechanism

On the LOESS grid every window contains exactly (20, 20) points, so unequal group sizes ? the

5. Corrected inference

5.1 E5 ? What survives on iPOP

Having established that the pipeline is not interpretable, the question the existing critique

![[iPOP under corrected inference](figures/fig5_corrected_ipop.png)]

*Figure 5. (a) What survives each of the three separated claims. (b) The single positive finding,

| Layer | Variables | C1 any age association (BH / BY) | C2 nonlinear | C2 discrete transition (BH / BY) | Power-equalised DE-SWAN

*(All p-values permutation-calibrated against a shuffle-before-modelling null, 200 permutations

The result is stark and, on reflection, unsurprising given §4. **iPOP contains almost no

The power-equalised DE-SWAN ? group sizes fixed at (k, k) so that the sample-size component of

![[Power-equalised DE-SWAN against its permutation null](figures/fig5b_power_equalised.png)]

The crest, reported honestly, is uninformative. The waves literature reports the crest age as a

| Layer | Published-style point estimate | Bootstrap median | 95 % CI | Width |

Four of six intervals span 20?24 years ? essentially the entire analysed range ? and cover both

5.2 E7/E7b ? Attacking the one positive finding

One cell of the table above is not null: **46/66 cytokine analytes nonlinear and 35/66 with a

Outlier diagnostics show the cytokine panel is unlike the others: median excess kurtosis

Rank-inverse-normal transform. Replacing values by Blom scores is monotone ? it preserves any

Leave-one-subject-out. Dropping the single subject aged 25.9 leaves 2.8 % of the median

The decisive control. Re-running the full RINT battery on nested subsets:

| Subset | n | Nonlinear | Discrete transition | Breakpoints |

Dropping a random subject leaves a median of 7 significant analytes; dropping that one subject

The one apparent transition in iPOP under corrected inference is a single 26-year-old with an